

Bayesian Learning Computer Lab 3

You are recommended to use R for solving the labs.

You work and submit your labs in pairs, but both of you should contribute equally and understand all parts of your solutions.

It is not allowed to share exact solutions with other student pairs.

The submitted lab reports will be verified through OURIGINAL and indications of plagiarism will be investigated by the Disciplinary Board.

Submit your solutions via LISAM, no later than May 19 at 23:59.

Please note the following about the format of the submitted lab report:

1. The lab report should include all solutions and plots to the stated problems with necessary comments.
2. Submit the lab report with your code attached to the solution of each sub-problem (1a), 1b),...) in **one** PDF document.
3. Submit a separate file containing all code.

1. Gibbs sampling for the logistic regression

Consider again the logistic regression model in problem 2 from the previous computer lab 2. Use the prior $\beta \sim \mathcal{N}(0, \tau^2 I)$, where $\tau = 3$.

- (a) Implement (code!) a Gibbs sampler that simulates from the **joint posterior** $p(\omega, \beta | \mathbf{x})$ by **augmenting** the data with **Polya-gamma latent variables**¹ $\omega_i, i = 1, \dots, n$. The full conditional posteriors are given on the slides from Lecture 7. Evaluate the convergence of the Gibbs sampler by calculating the Inefficiency Factors (IFs) and by plotting the trajectories of the sampled Markov chains.
- (b) Repeat the same task as in a), but now only use the first m observations of the dataset. Do this for all $m \in \{10, 40, 80\}$.

2. Metropolis Random Walk for Poisson regression

Consider the following Poisson regression model

$$y_i | \beta \stackrel{iid}{\sim} \text{Poisson} \left[\exp(\mathbf{x}_i^T \beta) \right], \quad i = 1, \dots, n,$$

where y_i is the count for the i th observation in the sample and $\mathbf{x}_i$ is the p -dimensional vector with covariate observations for the i th observation. Use the data set `eBayNumberOfBidderData_2025.dat`. This dataset contains observations from 700 eBay auctions of coins. The response variable is **nBids** and records the number of bids in each auction. The remaining variables are features/covariates (**x**):

¹Nicholas G. Polson, James G. Scott & Jesse Windle (2013) Bayesian Inference for Logistic Models Using Pólya–Gamma Latent Variables, Journal of the American Statistical Association

- **Const** (for the intercept)
 - **PowerSeller** (equal to 1 if the seller is selling large volumes on eBay)
 - **VerifyID** (equal to 1 if the seller is a verified seller by eBay)
 - **Sealed** (equal to 1 if the coin was sold in an unopened envelope)
 - **MinBlem** (equal to 1 if the coin has a minor defect)
 - **MajBlem** (equal to 1 if the coin has a major defect)
 - **LargNeg** (equal to 1 if the seller received a lot of negative feedback from customers)
 - **LogBook** (logarithm of the book value of the auctioned coin according to expert sellers. Standardized)
 - **MinBidShare** (ratio of the minimum selling price (starting price) to the book value. Standardized).
- (a) Obtain the maximum likelihood estimator of β in the Poisson regression model for the eBay data [Hint: `glm.R`, don't forget that `glm()` adds its own intercept so don't input the covariate `Const`]. Which covariates are significant?
- (b) Let's do a Bayesian analysis of the Poisson regression. Let the prior be $\beta \sim \mathcal{N}[\mathbf{0}, 100 \cdot (\mathbf{X}^T \mathbf{X})^{-1}]$, where $\mathbf{X}$ is the $n \times p$ covariate matrix. This is a commonly used prior, which is called Zellner's g-prior. Assume first that the posterior density is approximately multivariate normal:

$$\beta|y \sim \mathcal{N}(\tilde{\beta}, J_{\mathbf{y}}^{-1}(\tilde{\beta})),$$

where $\tilde{\beta}$ is the posterior mode and $J_{\mathbf{y}}(\tilde{\beta})$ is the negative Hessian at the posterior mode. $\tilde{\beta}$ and $J_{\mathbf{y}}(\tilde{\beta})$ can be obtained by numerical optimization (`optim.R`) exactly like you already did for the logistic regression in Lab 2 (but with the log posterior function replaced by the corresponding one for the Poisson model, which you have to code up.).

- (c) Let's simulate from the actual posterior of β using the Metropolis algorithm and compare the results with the approximate results in b). Program a general function that uses the Metropolis algorithm to generate random draws from an *arbitrary* posterior density. In order to show that it is a general function for any model, we denote the vector of model parameters by θ . Let the proposal density be the multivariate normal density mentioned in Lecture 8 (random walk Metropolis):

$$\theta_p|\theta^{(i-1)} \sim N(\theta^{(i-1)}, c \cdot \Sigma),$$

where $\Sigma = J_{\mathbf{y}}^{-1}(\tilde{\beta})$ was obtained in b). The value c is a tuning parameter and should be an input to your Metropolis function. The user of your Metropolis function should be able to supply her own posterior density function, not necessarily for the Poisson regression, and still be able to use your Metropolis function. This is not so straightforward, unless you have come across *function objects* in R. The note **HowToCodeRWM.pdf** in Lisam describes how you can do this in R.

Now, use your new Metropolis function to sample from the posterior of β in the Poisson regression for the eBay dataset. Assess MCMC convergence by graphical methods.

- (d) Use the MCMC draws from c) to simulate from the predictive distribution of the number of bidders in a new auction with the characteristics below. Plot the predictive distribution. What is the probability of no bidders in this new auction?

- **PowerSeller** = 1
- **VerifyID** = 0
- **Sealed** = 1
- **MinBlem** = 0
- **MajBlem** = 1
- **LargNeg** = 0
- **LogBook** = 1.3
- **MinBidShare** = 0.7

3. Time series models in Stan

- (a) Write a function in R that simulates data from the AR(1)-process

$$x_t = \mu + \phi(x_{t-1} - \mu) + \varepsilon_t, \quad \varepsilon_t \stackrel{iid}{\sim} N(0, \sigma^2),$$

for given values of μ , ϕ and σ^2 . Start the process at $x_1 = \mu$ and then simulate values for x_t for $t = 2, 3, \dots, T$ and return the vector $x_{1:T}$ containing all time points. Use $\mu = 5$, $\sigma^2 = 9$ and $T = 300$ and look at some different realizations (simulations) of $x_{1:T}$ for values of ϕ between -1 and 1 (this is the interval of ϕ where the AR(1)-process is stationary). Include a plot of at least one realization in the report. What effect does the value of ϕ have on $x_{1:T}$?

- (b) Use your function from a) to simulate two AR(1)-processes, $x_{1:T}$ with $\phi = 0.4$ and $y_{1:T}$ with $\phi = 0.98$. Now, treat your simulated vectors as synthetic data, and treat the values of μ , ϕ and σ^2 as unknown parameters. Implement Stan-code that samples from the posterior of the three parameters, using suitable non-informative priors of your choice. [Hint: Look at the time-series models examples in the Stan user's guide/reference manual, and note the different parameterization used here.]
- i. Report the posterior mean, 95% credible intervals and the number of effective posterior samples for the three inferred parameters for each of the simulated AR(1)-process. Are you able to estimate the true values?
 - ii. For each of the two data sets, evaluate the convergence of the samplers and plot the joint posterior of μ and ϕ . Comments?

GOOD LUCK!
BEST, BERTIL